

Day 12 Final Project Summary

Linear and Logistic Regression - notebooks, results, recommendations and coverage

Ruturaj Mokashi

Executive summary

Learning Day 12 focused on regression as a practical supervised learning toolkit. The project begins with interpretable linear regression, then moves into regularized regression, count-data GLMs, logistic regression classification, and advanced diagnostics. The final result is a five-notebook package that covers the MVD requirements and the additional professional topics agreed during planning.

Main conclusion

The full project is complete: all MVD tasks are covered, the topic-title gap around Logistic Regression is covered, and the advanced missing topics are captured in Notebook 5. The strongest business pattern is that model choice depends on the target type: continuous Sales, regularized numeric outcomes, counts, or classification probabilities.

Notebook package

Notebook	Main focus	Datasets	Core models
01 Linear Regression Advertising	Simple and multiple linear regression, OLS, residual diagnostics, statsmodels OLS.	Advertising	Linear Regression, OLS
02 Regularized Regression	Ridge, Lasso, ElasticNet, coefficient shrinkage, Lasso path, LassoCV.	Diabetes regression dataset	Ridge, Lasso, ElasticNet, LassoCV
03 GLM Count Models	Bike Sharing count prediction, Poisson, Negative Binomial, overdispersion.	Bike Sharing	Linear Regression, PoissonRegressor, Poisson GLM, Negative Binomial GLM
04 Logistic Regression	Binary classification, probability, thresholding, calibration and lift.	Customer campaign response	Logistic Regression
05 Advanced Diagnostics	Missing advanced topics and influence diagnostics.	Advertising, Bike Sharing, campaign response	OLS diagnostics, gradient descent, multinomial logistic regression

Key results

Notebook	Best/important result	Business meaning
Notebook 1	Multiple regression $R^2 = 0.8870$, RMSE = 1.6846; TV-only $R^2 = 0.6401$, RMSE = 3.0064.	Using TV, Radio and Newspaper together explains Sales much better than TV alone.
Notebook 2	LassoCV selected $\alpha = 1.6552$ with test RMSE = 52.9196 and Test $R^2 = 0.4714$.	Regularization improved stability and selected a simpler model by shrinking weak features.
Notebook 3	Linear Regression had best MAE = 8.194; Negative Binomial had better count-model deviance = 4.213.	For operational accuracy use the best MAE model; for count-model fit, NB is more statistically suitable when overdispersion exists.
Notebook 4	Baseline Logistic Regression accuracy = 0.777, ROC-AUC = 0.614, PR-AUC = 0.349.	The model ranks responders somewhat better than random but threshold tuning is required for campaign targeting.

Notebook	Best/important result	Business meaning
Notebook 5	Closed-form OLS matched statsmodels coefficients exactly; campaign multiclass model test accuracy = 0.700.	Diagnostics confirm how coefficients are learned and how to test model assumptions.

Project narrative

The project follows a clean analytical story. Notebook 1 establishes the foundation: how a regression line is fitted, what coefficients mean, and how residuals reveal the quality of predictions. Notebook 2 shows what happens when normal regression becomes too sensitive to features and how Ridge, Lasso and ElasticNet control this. Notebook 3 switches the target type from normal continuous values to counts, where Poisson and Negative Binomial models become more appropriate. Notebook 4 covers the classification side of the Day 12 title through Logistic Regression. Notebook 5 closes the advanced gaps: closed-form OLS, gradient descent, intervals, heteroscedasticity, influence, quasi-Poisson, and multinomial logistic regression.

Final recommendation

Use case	Recommended model family	Reason
Explain sales impact of marketing budgets	Multiple Linear Regression + statsmodels OLS	Interpretable coefficients, R ² , p-values and confidence intervals.
Many related numeric features	Ridge or ElasticNet	Stabilizes coefficients when features overlap.
Feature selection and simpler model	Lasso or LassoCV	Can set weak coefficients to zero.
Count outcome with overdispersion	Negative Binomial GLM	Handles variance larger than mean better than Poisson.
Binary response or churn-style outcome	Logistic Regression	Outputs probabilities, supports threshold tuning and lift analysis.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.